Istabrak Abbes

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EDUCATION

Université de Montréal/Mila — Montréal, Canada
Doctor of Philosophy - Computer Science (Supervisor: Dr. Sarath Chandar) — *Aug 2025 – Present*

Université de Montréal/Mila — Montréal, Canada
Master of Computer Science (Supervisor: Dr. Sarath Chandar) — *Aug 2023 – May 2025*
GPA 3.67/4.0 : Fast tracked to PhD

National Engineering School of Tunis (ENIT) — Tunis, Tunisia
Bachelors of Technology in Computer Science — *Aug 2019 – May 2022*

EXPERIENCE

ServiceNow Research – CoreAI — Montreal, Canada
Research Intern, Agent Routing — *August 2026 – Present*

- Developing an agent router that selects models turn by turn to reduce inference cost while preserving task success.
- Designing trace-driven routing methods that use execution failures and task difficulty to choose model size and reasoning effort.

Mila - Quebec AI Institute — Montreal, Canada
Graduate Researcher — *September 2023 – Present*

- **Projects** Optimizing the continual pretraining process of large language models on HPC, on Groundedness detection
- **Keywords** Continual Learning, Catastrophic Forgetting, Language Models, Hallucination Detection, PyTorch, Distributed Training, High-performance computers (Frontier)

Sofrecom (Part of the Orange Group) — Tunisia
Data Governance Manager — *December 2022 – May 2023*

- **Tasks** Managed data governance, data quality, data catalog, and data lineage projects along with extensive data analysis.
- **Keywords** Data Governance, Data Quality, Data Analysis, Deep Learning, NLP, PyTorch, Dataiku DSS

Mila - Quebec AI Institute — Montreal, Canada
Deep Learning Research Intern — *July – October 2021*

- **Project** Developed a clustering method for the memory buffer used by the critical gradient optimizer.
- **Keywords** Deep Learning, Critical Gradient Optimizer, Continual Learning, Optimization, PyTorch

Sofrecom (Part of the Orange Group) — Tunisia
Deep Learning Research Intern — *February – June 2022*

- **Projects** Built a Voice-Bot for skills assessment using reverse chatbot logic where the system asks questions and evaluates responses. Also developed a full-stack web application and implemented a face recognition model.
- **Keywords** RASA, MongoDB, Angular, TensorFlow.js, Flask, Azure, Web Scraping, CamemBERT, DIET, Transformers

National Engineering School of Tunis — Tunisia
Data Science Intern — *August – October 2021*

- **Project** Scraped Instagram data (likes, comments, photos) to study factors influencing tourist site visits. Developed classifiers and a pipeline to detect the presence of famous personalities in images.
- **Keywords** Instagram Scraping, Selenium, BeautifulSoup, ResNet, Transfer Learning

Arion Technologies — Tunisia
Data Science Intern — *June – August 2021*

- **Project** Implemented a deep learning recommendation system using the GRU4Rec architecture tailored for retail applications.
- **Keywords** RNN, GRU, GRU4Rec, Collaborative Filtering, Azure

Selected Papers

Forgotten or Inaccessible? Restoration Savings Reveal What Language Models Retain After Fine-Tuning 2026
– **Istabrak Abbes**, Irina Rish, Sarath Chandar
Submitted to ICLR 2027

Failed Reasoning Traces Tell You What Is Fixable (But Not by Reading Them) 2026
– Nizar Islah, **Istabrak Abbes**, Irina Rish, Sarath Chandar, Eilif B Muller
Submitted to COLING 2027

What Does Layer-Importance Reveal About Transformers and State-Space Models? 2026
– **Istabrak Abbes**, Nizar Islah , Irina Rish, Sarath Chandar
Empirical Methods in Natural Language Processing (EMNLP)

Pretraining Large Language Models with Latent Reinforcement Learning 2026
– **Istabrak Abbes***, Gopeshh Subbaraj, Artem Zholus, Matthew Riemer, Irina Rish, Sarath Chandar
Workshop at ICLR 2026

Revisiting Replay and Gradient Alignment for Continual Pretraining of Large Language Models 2025
– **Istabrak Abbes***, Gopeshh Subbaraj*, Benjamin Therien, Matthew Riemer, Sarath Chandar, Irina Rish
Conference on Lifelong Learning Agents (CoLLAs)

Small Encoders Can Rival Large Decoders in Detecting Groundedness 2025
– **Istabrak Abbes***, Gabriel Prato, Quentin Fournier, Adam Elwood, Fernando Rodriguez, Sarath Chandar
Association for Computational Linguistics (ACL)

Context-aware Language Modeling for Arabic Misogyny Identification 2021
– **Istabrak Abbes***, Eya Nakache, Moez BenHajhmida
Forum for Information Retrieval Evaluation (FIRE)